EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Li-6800 DATA

- Importing stomatal conductance data obtained from Li-6800
- There are individual spreadsheets for each plant sampled with the Li-6800
- Reps 1, 2 and 3 were used for measurements. Rep 4 was not used for Li-6800 measurements
- There are a total of 12 individual Li-6800 spreadsheets
- Each spreadsheet contains a 'Raw Data' tab and a 'Sorted Data' tab. The 'Sorted Data' tab is the one to use for analyses.
- The key data parameters to assess are 'gsw' (which we can refer to as 'gs') (Column P) and 'A' (net photosynthetic carbon assimilation) (Column K)
- The light level (to determine if it is phase 1 dark, phase 2 light or phase 3 dark) is denoted in column DG ### Load the libraries

```
library(knitr)
#knitr::spin("Exp801.li6800.import.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/")
library(dae)
library(dplyr)
library(growthPheno) # calcLagged
packageVersion("growthPheno")
```

```
[1] '3.1.18'
```

Set up constants and files

```
options(width=90)
inputFiles <- list.files("../data")
inputFiles <- inputFiles[grepl("L01_", inputFiles)]
writeFile  <- "../data/li6800.v1.Rdata"
```

Function for reading in each of the Li-6800 files

```
readli6800files <- function(file)
{
  tbl <- suppressMessages(readxl::read_xlsx(paste0("../data/", file),
                                            sheet = "Sorted Data",
                                            col_names = FALSE)[,c(1:6, 11, 16, 21, (26*4+7))]) ##column



  ##tidy up
  colnames(tbl) <- c("obs", "time", "elapsed","date","hhmmss","time.elapsed", "A",
                     "gsw", "TleafEB", "light")
  tbl <- tbl %>% slice(-c(1:3))

  ##convert variables
  tbl <- tbl %>% mutate(across(c("A", "gsw", "TleafEB", "light"), as.numeric)) %>% as.data.frame()
  tbl$Timestamp <- as.POSIXct(tbl$date, format = "%Y%m%d %H:%M:%S")

  ##calculate time difference between each observation
```

```
  tbl$time.diff <- c(0, diff(tbl$Timestamp))
  tbl$xElapsed <- cumsum(tbl$time.diff)
  tbl$xElapsed.min <- tbl$xElapsed/60

  ##append treatment information using filename
  trt.info <- strsplit(file, "_")
  tbl$Line <- trt.info[[1]][1]
  tbl$Temperature <- trt.info[[1]][2]
  tbl$Watering <- trt.info[[1]][3]
  tbl$Rep <- trt.info[[1]][4]

  return(tbl)
}

li6800.dat <- do.call(rbind, lapply(inputFiles, readli6800files))
head(li6800.dat)
```

```
  obs         time elapsed              date   hhmmss        time.elapsed            A
1   2 1750028474.5       1 20250616 09:01:14 09:01:14                   0 20.534477055
2   3 1750028504.5      31 20250616 09:01:44 09:01:44 0.49999999999999822 -1.155007249
3   4 1750028534.5      61 20250616 09:02:14 09:02:14 0.99999999999999645  0.437746106
4   5 1750028564.5      91 20250616 09:02:44 09:02:44  1.4999999999999947  0.494324600
5   6 1750028594.5     121 20250616 09:03:14 09:03:14  1.9999999999999929  0.088890523
6   7 1750028624.5     151 20250616 09:03:44 09:03:44  2.4999999999999911  0.005596654
        gsw  TleafEB        light           Timestamp time.diff xElapsed xElapsed.min
1 0.1826104 24.47397 999.96293103 2025-06-16 09:01:14         0        0           0.0
2 0.1551714 23.50401   0.00500015 2025-06-16 09:01:44        30       30           0.5
3 0.1721729 23.26821   0.00500015 2025-06-16 09:02:14        30       60           1.0
4 0.1564419 23.22260   0.00500015 2025-06-16 09:02:44        30       90           1.5
5 0.1273621 23.26191   0.00500015 2025-06-16 09:03:14        30      120           2.0
6 0.1033265 23.30610   0.00500015 2025-06-16 09:03:44        30      150           2.5
  Line Temperature Watering Rep
1  L01         Amb      Lim Re1
2  L01         Amb      Lim Re1
3  L01         Amb      Lim Re1
4  L01         Amb      Lim Re1
5  L01         Amb      Lim Re1
6  L01         Amb      Lim Re1
```

```
nrow(li6800.dat) #4392
```

```
[1] 4392
```

```
li6800.dat$Rep <- factor(li6800.dat$Rep)
li6800.dat$Rep <- dae::fac.recast(li6800.dat$Rep,
                                  newlevels = c(1:3))

##put together a unique identifier for each dataset (day)
li6800.dat$TWR <- paste(li6800.dat$Temperature, li6800.dat$Watering, li6800.dat$Rep, sep ="_")
```

How many measurements on each day (for each TWR)?

```
table(li6800.dat$TWR)
```

```
   Amb_Lim_1    Amb_Lim_2    Amb_Lim_3  Amb_Unlim_1  Amb_Unlim_2  Amb_Unlim_3
         366          366          366          366          366          366
  Heat_Lim_1   Heat_Lim_2   Heat_Lim_3 Heat_Unlim_1 Heat_Unlim_2 Heat_Unlim_3
         366          366          366          366          366          366
```

Check if the lights were off for the first measurement? If so, set A and gs to NA

```
li6800.dat %>% filter(xElapsed < 60 & light > 1) %>% select("TWR", "time.diff", "light")
```

```
            TWR time.diff     light
1     Amb_Lim_1         0  999.9629
2     Amb_Lim_2         0 1000.0082
3     Amb_Lim_3         0 1000.0089
4   Amb_Unlim_1         0 1000.0224
5   Amb_Unlim_2         0 1000.0129
6   Amb_Unlim_3         0  999.3537
7    Heat_Lim_1         0  999.9886
8    Heat_Lim_2         0  999.9988
9    Heat_Lim_3         0 1000.0091
10 Heat_Unlim_1         0  999.9835
11 Heat_Unlim_2         0 1000.0141
12 Heat_Unlim_3         0  999.9863
```

```
##this confirms that the light was only on for the first measurement of some treatments

li6800.dat[(li6800.dat$xElapsed == 0 & li6800.dat$light > 1 & !is.na(li6800.dat$A)),
           c("A", "gsw")] <- NA
```

Add location of the plant that was sampled

This information orginally came from the Porometer data spreadsheet, but I have created my own little spreadsheet to import here

```
plant.sampled.df <- read.csv("../data/PlantPositionsLookupTable.csv")
##remove those plants that have data used for analysis as "No", as this was not supplied to us
plant.sampled.df <- plant.sampled.df[plant.sampled.df$Data.Used.for.Analyses. == "Yes",]

plant.sampled.df <- within(plant.sampled.df,{
  Temperature <- Room
  Temperature <- factor(Temperature, levels = c("Ambient", "Heat"),
                        labels = c("Amb", "Heat"))
  Watering <- factor(Watering, levels = c("Limited", "Unlimited"),
                        labels = c("Lim", "Unlim"))
  Rep <- factor(Rep)
})

li6800.dat <- left_join(li6800.dat, plant.sampled.df[, c("Rep", "Watering", "Temperature", "Location")])
```

```
Joining with `by = join_by(Temperature, Watering, Rep)`
```

```
nrow(li6800.dat)
```

```
[1] 4392
```

Define the light and dark regions

```
hist(li6800.dat$light)
```

```
li6800.dat$Light <- "Off"
li6800.dat$Light[li6800.dat$light > 1] <- "On"
table(li6800.dat$TWR, li6800.dat$Light)
```

```
              Off  On
  Amb_Lim_1    243 123
  Amb_Lim_2    244 122
  Amb_Lim_3    243 123
  Amb_Unlim_1  243 123
  Amb_Unlim_2  243 123
  Amb_Unlim_3  243 123
  Heat_Lim_1   244 122
  Heat_Lim_2   243 123
  Heat_Lim_3   243 123
  Heat_Unlim_1 243 123
  Heat_Unlim_2 243 123
  Heat_Unlim_3 243 123
```

Find anomalies in gsw

set the value to NA

```
li6800.dat[li6800.dat$gsw < -0.01 & !is.na(li6800.dat$gsw),]
```

```
   obs               time           elapsed               date   hhmmss       time.elapsed
74  75 1750030663.5999999 2190.099999904633 20250616 09:37:43 09:37:43 36.483333333333334
          A        gsw  TleafEB      light           Timestamp time.diff xElapsed
74 1.540777 -0.1121717 23.90939 0.00500015 2025-06-16 09:37:43        30     2189
   xElapsed.min Line Temperature Watering Rep       TWR Location Light
74     36.48333  L01         Amb      Lim   1 Amb_Lim_1      D13   Off
```

```
li6800.dat$gsw[li6800.dat$gsw < -0.01] <- NA
```

Visualise the A and gsw data

```
plotProfiles(li6800.dat, response = "A",
             individuals = "Location", times = "xElapsed.min",
             facet.x = "Temperature", facet.y = "Watering",  colour.column = "TWR",
             y.title = "A",
             x.title = "Seconds",
             title = "A",
             breaks.spacing.x = 30,
             alpha = 0.75)
```

```
plotProfiles(li6800.dat, response = "gsw",
             individuals = "Location", times = "xElapsed.min",
             facet.x = "Temperature", facet.y = "Watering",  colour.column = "TWR",
             y.title = "gsw",
             x.title = "Seconds",
             title = "gsw",
```

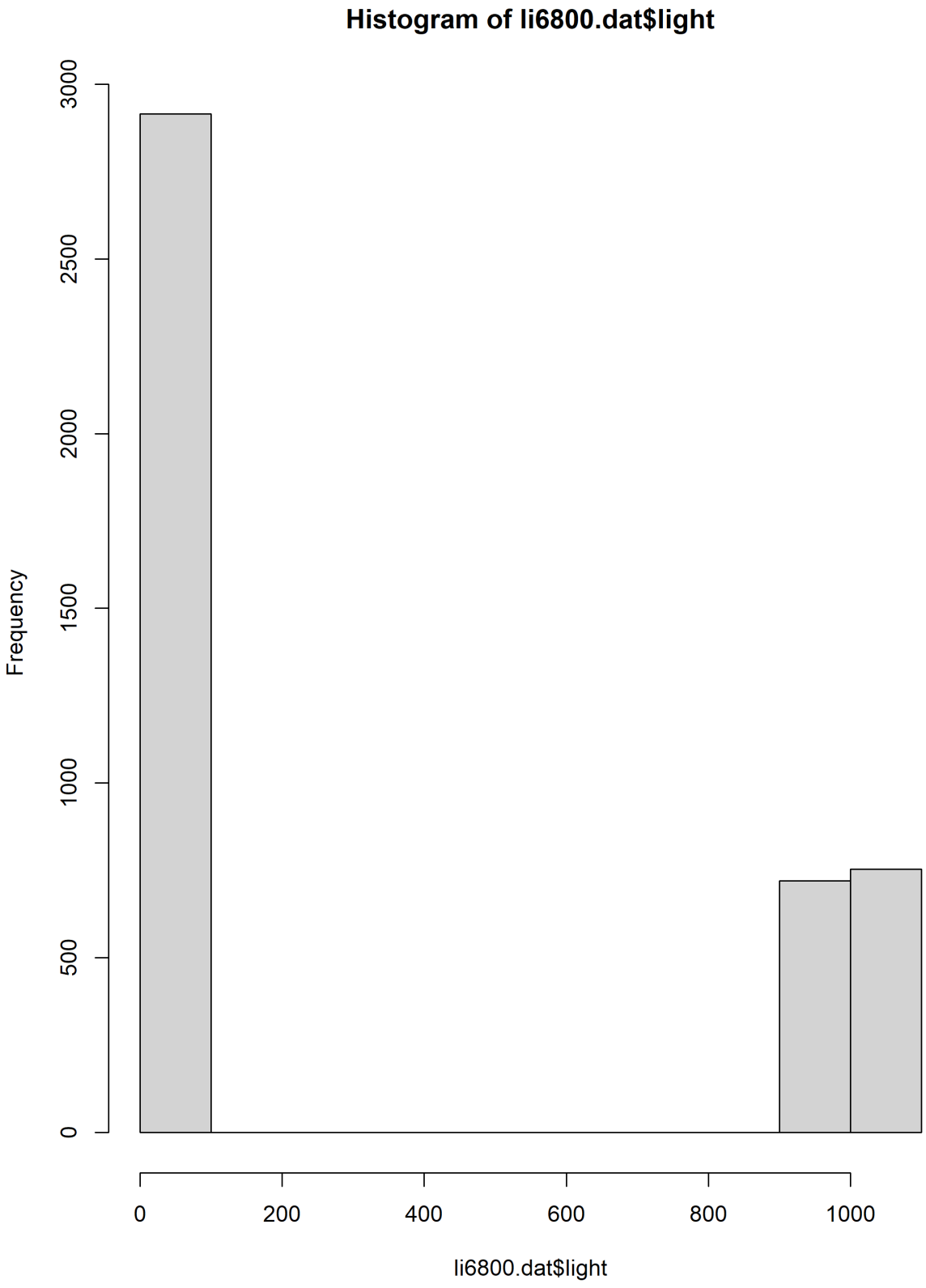


Figure 1: plot of chunk unnamed-chunk-7

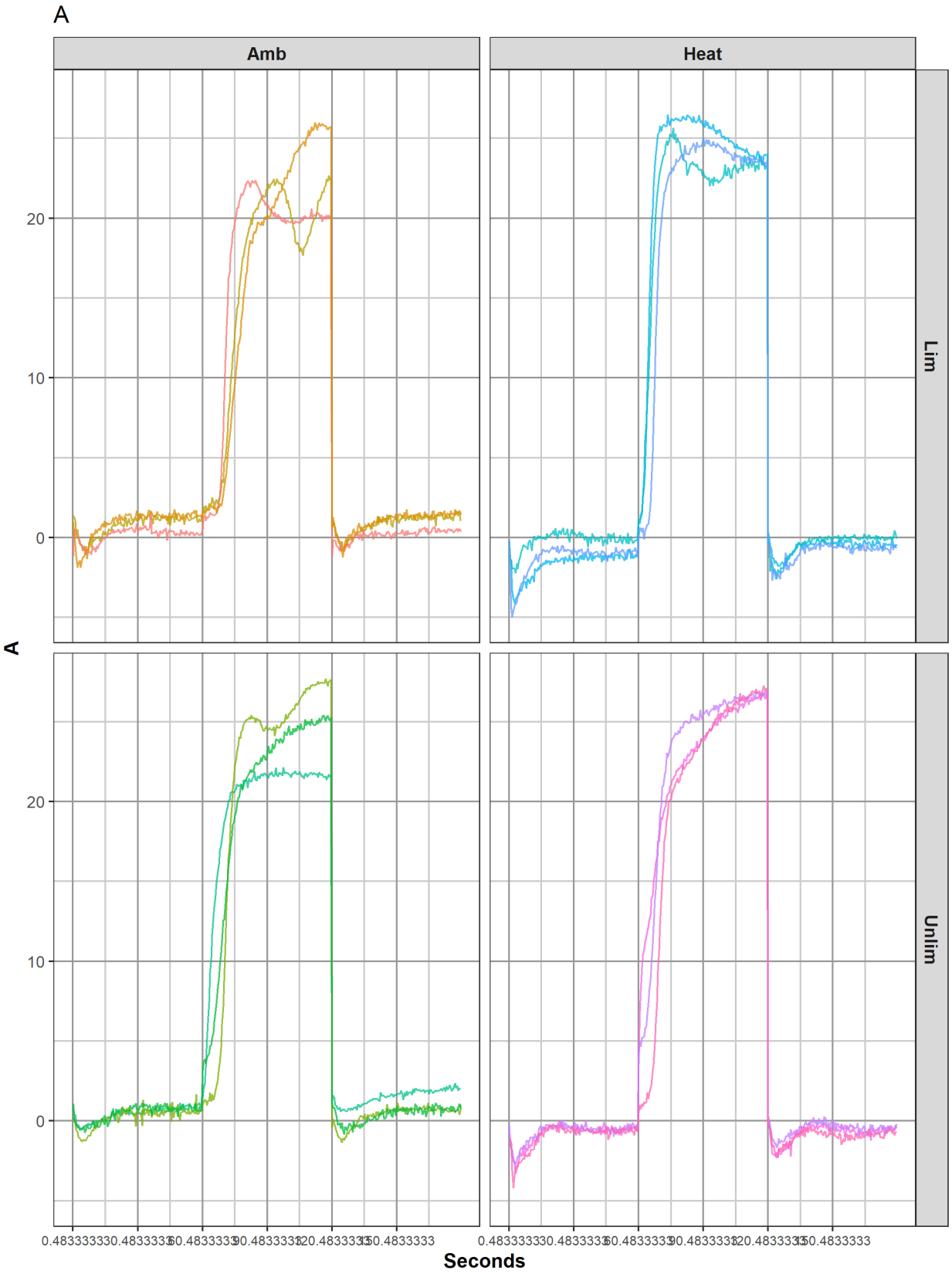


Figure 2: plot of chunk unnamed-chunk-9

```
            breaks.spacing.x = 30,
            alpha = 0.75)
```

```
save.image(writeFile)
```

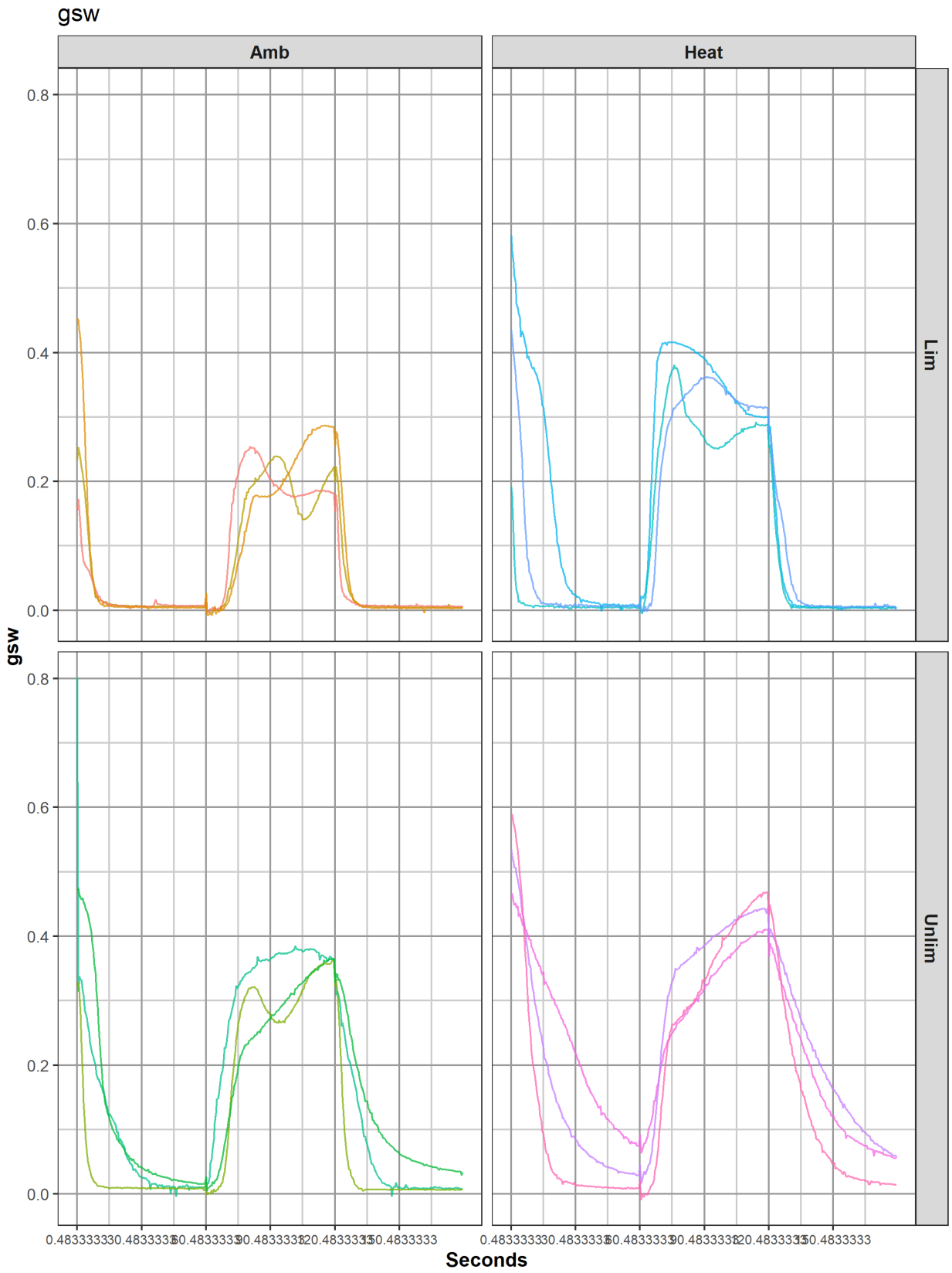


Figure 3: plot of chunk unnamed-chunk-9